

Enhancing Emotion Recognition in Text with Stacked CNN-BiLSTM Framework

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Abstract—Emotions play a significant impact on shaping the human experience, and humans tend to express emotions in all forms of communication, including text, especially in the social media era. Recognizing emotions in text is crucial for various applications, enabling a better understanding and the design of emotion-aware responses. This study proposes a novel framework for textual emotion recognition using a stacked CNN-BiLSTM. This framework aims to leverage the advantages of both the algorithms effectively, especially in the context of the increasingly intricate conversation found in social media. Further, the efficacy of the framework with hybrid embedding methods is demonstrated in the experiments using various optimizers and activation functions. The promising results validate the application of this novel approach for inherently complex input texts.

Keywords—Emotion Recognition, Word2Vec, Convolutional Neural Network, FastText, Bidirectional Long Short Term Memory.

I. INTRODUCTION

Emotion delineates particular sensation that characterizes an individual's psychological state. Emotions are subjective, as they depend on the individual's experience, circumstances, or other external factors. They have a significant impact on beliefs, actions, and general well-being, shaping the human experience. Humans express their emotions through speech, facial expression, text, and gestures. The advent of social media platforms has provided individuals with a forum for sharing and expressing their ideas. This information can be used by analysts to monitor people's opinions regarding their products or to identify the emotional state of individuals. Sentiment analysis doesn't capture the intricate emotions of the individuals or the degree of their response [1]; it merely tells whether people are reacting positively, neutrally, or negatively.

Emotion Recognition in text aims to identify emotions namely joy, anger, fear, surprise, sadness, etc. [2]. It is challenging due to its complex nature, including issues such as polysemy, spelling mistakes, or implied emotions. Researchers have recently put forth several kinds of approaches, including learning-based, lexical affinity, hybrid, and keyword-based models, for recognizing text's emotions [3]. First, they presented a rule-based method

that comprised two methods viz: keyword-based, lexical affinity-based. Subsequently, the learning-based technique emerged as a novel approach. In an attempt to achieve high accuracy, many researchers are also currently integrating the methodologies and developing hybrid models [4].

This paper offers a distinctive method for textual emotion recognition utilizing a Stacked Convolution Neural Network-Bidirectional Long Short-Term Memory (CNN-BiLSTM) [5] framework. The document's remaining section is organized as: Section 2 is a review of the works on emotion recognition. A description of the data collection and suggested framework technique can be found in Section 3. The findings and analysis of the proposed work will be discussed in Section 4, which is the next section. Lastly, Section 5 delivers the paper's conclusion.

II. RELATED WORKS

Different authors have put forward proposals for subsequent steps in emotion detection, such as creating embedding and using Machine Learning (ML) or Deep Learning (DL) models.

One such idea leverages DL models, like CNN and LSTM [6], to identify emotions from Twitter comments. Following a comparative study of the models' performances, the researchers found that LSTM models performed better than CNN in terms of training time and F1 score. This highlights that LSTM-based models are better suited of capturing contextual information. In another research project [7], various learning models are compared, and an architecture that involves stacking numerous BiLSTM layers to evaluate performance has been proposed.

Work [8] and [9] explores the relationship between the reliability of news and its underlying emotions, which further develops a multitask framework which incorporates multiclass emotion categorization as an additional work for identifying rumours and false information.

The work [10] focuses on sentiments and emotion identification on Twitter data leveraging the NLTK and TextBlob modules. The study emphasizes the value of opinion mining on social media sites and how it can be used to better understand the views and mindsets of the wider population. According to the survey papers [11] and [12] the accuracy of the emotion analysis system is dependent on the accuracy achieved at each stage. DL models likely to perform better than ML methodologies, in accordance with empirical findings. For the aforementioned reason, the technique that utilizes deep learning to extract the underlying emotions from text byutilizing CNN and BiLSTMwas developed.

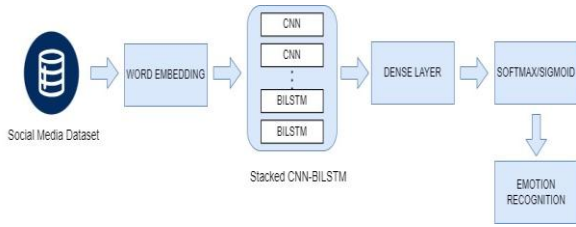


Fig. 1. Proposed stacked CNN-BiLSTM Framework

III. METHODOLOGY

This section is divided into several stages, such as Data collection and pre-processing, word embedding, model architecture and discussion.

Data Collection and Pre-processing

As a part of data collection, the Google API was utilized to collect comments from YouTube videos relevant to the research topic as part of data gathering and preparation approach. Further, pre-processing was also done, with tasks such as lemmatization, tokenization, punctuations, etc. The pre-processed data was labelled by annotators into six different emotion categories viz: love, fear, joy, sadness, anger, and surprise.

Word Embeddings

1.Word2Vec:Word2Vec[13] is a popular word embedding technology. In order to learn word representations depending on their surrounding context, it uses neural network. The two main models offered by Word2Vec include Continuous Bag of Words (CBOW) and Skip-Gram. The former estimates the target word given its context, whereas the latter forecast the words' context given anintended word [14]. This embedding method has demonstrated efficacy in capturing the semantic and syntactic connections among words.

2.FastText: By adding subword information, FastText expands the concept of word embeddings. FastText divides words into smaller subword units called character n-grams rather than representing words as entire words. With this method, it can handle morphological variants and provides embeddings for words that are not in its vocabulary as well. FastText is especially helpful in

managing languages with complex morphologies and also in enhancing the visualization of uncommon or obscure terms [15].

An integrated word embedding with FastText and Word2Vec has been proposed. Word2Vec is superior at capturing semantic relationships, whereas FastText is better at managing unusual words and subword information. This integrated word embedding model create embeddings that can more accurately describe language's subtleties and provide words with a complete representation. The combined method is particularly useful in situations when a broad vocabulary and strong semantic comprehension are necessary especially in multilingual environments, applications with domain-specific jargon, etc.

Fig 1. depicts the general architecture of the framework. Thereare four primary steps in the approach viz: Word Embedding, Stacked CNN-BiLSTM, Dense layer, and activation function. Initially, through word embedding, the syntactic and semantic characteristics of words and phrases are captured as numerical vectors from the raw text data. Further, to extract complicated patterns and sequences from the word embeddings, Stacked CNN-BiLSTM that uses a hybrid neural network architecture is employed. The dimension of the output of the stacked CNN-BiLSTM is reduced by the dense layer, before the last classification stage. To figure out the emotion connected to the input data, the dense layer's output is fed into the activation layer.

The Proposed Stacked CNN-BiLSTM Framework describes a hybrid neural network design with convolution and BiLSTM layers that is specifically designed for sequence classification. Three 1D convolutional layers with progressively larger filters and ReLU activation make up the base of the model. After every convolutional layer, batch normalization is used to ensure consistent training. In the third convolutional layer, a crucial dropout layer is included for better regularization. Subsequent convolutional blocks that employ batch normalization and progressively bigger filter sizes expand on this concept. The last two blocks incorporate dropout layers with caution. After a substantial 1D convolutional layer with 128 filters, bidirectional LSTM layers are added. The 128-unit first bidirectional LSTM layer is expanded with dropout layers.

An additional bidirectional LSTM layer of 256 units, followed by a dropout layer, substantially improves the model's ability to capture complex sequential patterns. The architecture has been wrapped up with a final 128-unit BiLSTM layer. The dense layer—which has six units and activation—is crucial for multi-class categorization. Due to its compilation strategy, which uses sparse categorical cross entropy loss, the optimizer adopting

learning rate - 0.001, the model is well-suited for tasks involving sequential data.

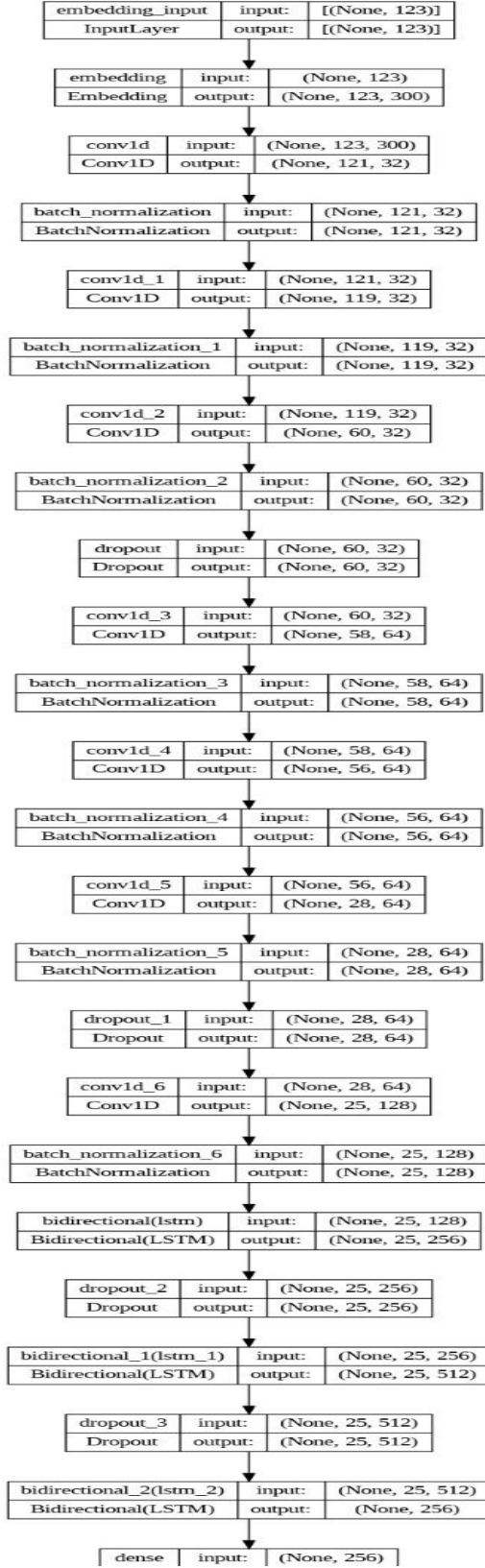


Fig 2. Design of Stacked CNN-BiLSTM

Fig.2. described the flow and design of the Stacked CNN-BiLSTM model. Furthermore, by tuning the activation functions and optimizers, the model's performance was further examined and optimized.

IV. RESULTS AND DISCUSSIONS

The GPU and Google Colab Environment are put to work in the experiment setting. In the current study, the higher-level Keras API Model was utilized.

Methodical experiments were conducted with different combinations of activation functions and optimizers to record the efficacy of the suggested model. The efficiency was assessed using various criteria encompassing recall, accuracy, precision, and F1 score.

TABLE 1 PERFORMANCE ANALYSIS OF STACKED CNN-BiLSTM WITH VARIOUS OPTIMIZERS AND ACTIVATION FUNCTION

Optimizer	Activation function	Precision	Recall	F1-Score	Accuracy
ADAM	Softmax	0.96	0.96	0.94	96%
	Sigmoid	0.93	0.94	0.93	95%
RMSprop	Softmax	0.94	0.94	0.94	95%
	Sigmoid	0.92	0.91	0.92	94%
SGD	Softmax	0.44	0.61	0.49	69%
	Sigmoid	0.41	0.59	0.48	61%

In order to capture the complex correlations found in the self-build social media dataset, Table 1 showed how well the suggested model with Softmax and sigmoid activation function with Adam [16], RMSprop, and SGD optimizer performed. The outcomes demonstrate that the highest accuracy of 96% is produced by the Adam optimizer using the Softmax function. The model's 94% precision score further emphasizes its ability to generate accurate predictions with few false positives.

The trials also evaluated how effectively the proposed model performed with the RMSprop optimizer when sigmoid and Softmax activation functions yielding respective competitive accuracy scores of 94% and 95%. Despite having a slightly lower accuracy, the sigmoid activation function had good precision at 92%, indicating its effectiveness in predicting true positives.

Notably, when the Stochastic Gradient Descent (SGD) [17] optimizer combined with the Softmax activation function, its accuracy and precision scores decreased by 69% and 44%, respectively. The accuracy and precision scores of the sigmoid activation function with SGD were significantly lower, at 61% and 41%, indicating how sensitive the model's effectiveness is to the choice of optimizer and activation function.

The model's performance across 50 epochs, trained with the Optimizer-Adam and activation function-Softmax, is

displayed in the Fig 3. The positive performance of the model can be observed from the fact that by 10 epochs the training and validation accuracy increased exponentially and after that stabilizes during the further epochs. The same can be correlated with the corresponding decrease in the training and validation loss as displayed on the right side of Fig 3.

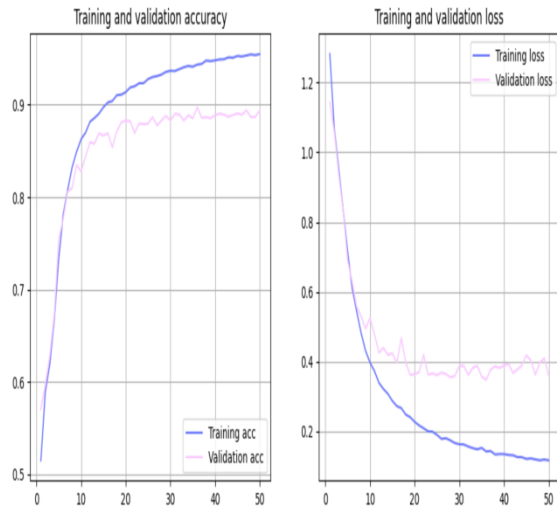


Fig 3. Performance graph with Adam-Softmax

Similar behaviour can be noticed in the model was trained with Optimizer-Adam and activation function-sigmoid as exemplified in the Fig 4.

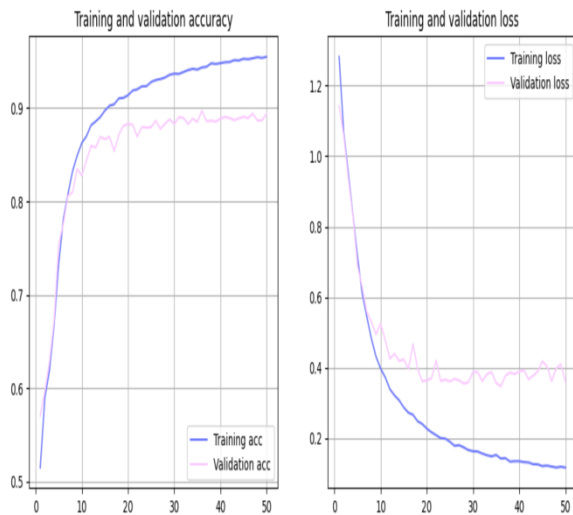


Fig 4. Performance graph with Adam-sigmoid

As shown in confusion matrix in Fig 5 it can be observed that for most of the input data the correct classes have been recognized. Also it can be seen that no anger category has been recognized as surprise and no fear category has been recognized as love. Only one instance of love category has been recognized as fear and surprise. There is only one recognized case of the surprise category that is love.

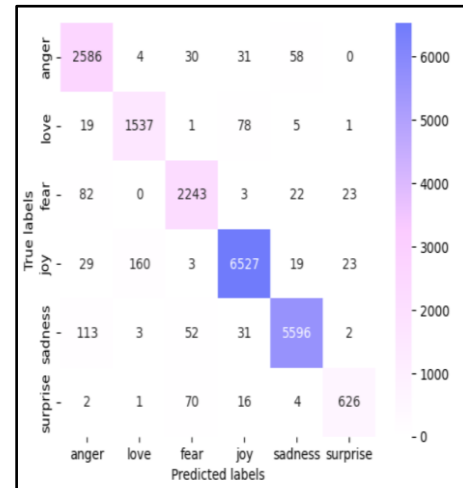


Fig. 5. Stacked CNN-BiLSTM framework's confusion matrix for RMSProp and Softmax

As shown in the confusion matrix in Fig 6 it can be observed that for most of the input data the correct classes have been recognized. Also it can be seen that no sadness category has been recognized as surprise and only one instance of surprise category has been recognized as sadness.



Fig. 6. Stacked CNN-BiLSTM framework's confusion matrix with RMSprop and sigmoid

V. CONCLUSIONS

A novel textual emotion recognition framework with Stacked CNN-BiLSTM was devised and the efficacy has been demonstrated with multiple optimizers and activation functions. It may be noted that a hybrid word embedding method was explored in this proposed framework. The social media dataset was employed to assess the effectiveness of the proposed framework, and the encouraging outcomes of multiple evaluations were reviewed and debated.

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REFERENCES

- [1] Li, Y., Lin, Z., Lin, Y. et al. Learning Sentiment-Enhanced Word Representations by Fusing External Hybrid Sentiment Knowledge. *Cogn Comput* 15, 1973–1987 (2023). <https://doi.org/10.1007/s12559-023-10164-1>
- [2] M. N. Alrasheedy, R. C. Muniyandi and F. Fauzi, "Text-Based Emotion Detection and Applications: A Literature Review," 2022 International Conference on Cyber Resilience (ICCR), Dubai, United Arab Emirates, 2022, pp. 1-9, doi: 10.1109/ICCR56254.2022.9995902
- [3] Alswaidan, Nourah and Mohamed El Bachir Menai. "A survey of state-of-the-art approaches for emotion recognition in text." *Knowledge and Information Systems* 62 (2020): 2937 - 2987.
- [4] Bharti SK, Varadhaganapathy S, Gupta RK, Shukla PK, Bouye M, Hingaa SK, Mahmoud A. Text-Based Emotion Recognition Using Deep Learning Approach. *Comput Intell Neurosci*. 2022 Aug 23;2022:2645381. doi:10.1155/2022/2645381. PMID: 36052029; PMCID: PMC9427219.
- [5] Rhanoui M, Mikram M, Yousfi S, Barzali S. A CNN-BiLSTM Model for Document-Level Sentiment Analysis. *Machine Learning and Knowledge Extraction*. 2019; 1(3):832-847. <https://doi.org/10.3390/make1030048>
- [6] K. H. Mahfouz and G. Al-Naymat, "Emotion Recognition from Twitter Comments Using Deep Learning," 2023 24th International Arab Conference on Information Technology (ACIT), Ajman, United Arab Emirates, 2023, pp. 1-5, doi: 10.1109/ACIT58888.2023.10453708.
- [7] Amit Pimpalkar, Jeberson Retna Raj R, MBiLSTM GloVe: Embedding GloVe knowledge into the corpus using multi-layer BiLSTM deep learning model for social media sentiment analysis, *Expert Systems with Applications*, Volume 203, 2022, 117581, ISSN 0957-4174
- [8] A. Choudhry, I. Khatri, M. Jain and D. K. Vishwakarma, "An Emotion-Aware Multitask Approach to Fake News and Rumor Detection Using Transfer Learning," in *IEEE Transactions on Computational Social Systems*, vol. 11, no. 1, pp. 588-599, Feb. 2024, doi: 10.1109/TCSS.2022.3228312.
- [9] Q. Guo, Z. Kang, L. Tian and Z. Chen, "TieFake: Title-Text Similarity and Emotion-Aware Fake News Detection," 2023 International Joint Conference on Neural Networks (IJCNN), Gold Coast, Australia, 2023, pp. 1-7, doi: 10.1109/IJCNN54540.2023.10191858.
- [10] Nehal, D. Jeet, V. Sharma, S. Mishra, C. Iwendi and J. Osamor, "Twitter Sentiment Analysis and Emotion Detection Using NLTK and TextBlob," 2023 4th International Conference on Computation, Automation and Knowledge Management (ICCAKM), Dubai, United Arab Emirates, 2023, pp. 1-6, doi: 10.1109/ICCAKM58659.2023.10449540.
- [11] D. Shukla and S. K. Dwivedi, "A Comparative Study of Text-Based Emotion Detection Techniques for Emotion Recognition on Social Media Data," 2023 IEEE 7th Conference on Information and Communication Technology (CICT), Jabalpur, India, 2023, pp. 1-6, doi: 10.1109/CICT59886.2023.10455381.
- [12] A. A. Maruf, F. Khanam, M. M. Haque, Z. M. Jiyad, M. F. Mridha and Z. Aung, "Challenges and Opportunities of Text-Based Emotion Detection: A Survey," in *IEEE Access*, vol. 12, pp. 18416-18450, 2024, doi: 10.1109/ACCESS.2024.3356357.
- [13] N. W. Method et al., "word2vec Explained: Deriving Mikolov et al," arXiv:1402.3722 [cs, stat], no. 2, 2014.
- [14] Mikolov, Tomas et al. "Distributed Representations of Words and Phrases and their Compositionality." *Neural Information Processing Systems* (2013).
- [15] Bojanowski, Piotr et al. "Enriching Word Vectors with Subword Information." *Transactions of the Association for Computational Linguistics* 5 (2016): 135-146.
- [16] Kingma, Diederik P. and Jimmy Ba. "Adam: A Method for Stochastic Optimization." *CoRR abs/1412.6980* (2014): n. pag.
- [17] Aatila, Mustapha et al. "An Overview of Gradient Descent Algorithm Optimization in Machine Learning: Application in the Ophthalmology Field." *International Conference on Smart Applications and Data Analysis* (2020).